Available online at www.sciencedirect.com

ScienceDirect

Current Opinion in Cell Biology

Telomeres and aging

Jerry W Shay

Telomeres (the TTAGGG repetitive DNA at the ends of linear chromosomes) are part of the 3D spatial organization of the nuclear genome. Long-range 3D chromatin interactions also establish specific patterns of regulated gene expression. An emerging area of interest is the role of telomere 3D looping with interstitial telomeric sequences (ITS) through interactions with telomere shelterin proteins. Telomeres form interstitial telomere loops (ITL) that interact with ITS and modify gene expression at distal genomic regions. Human laminopathies and telomeropathies often correlate with short telomeres. Since telomeres progressively shorten with increased turnover and chronological age in dividing somatic cells, ITLs may also change and have functional roles in normal and pathophysiological processes. Overall, telomeres help stabilize the nuclear genome with high fidelity throughout early adult life but diminish in post-reproductive age-associated pathology.

Address
Department of Cell Biology, UT Southwestern Medical Center, 5323 Harry Hines Blvd., Dallas, TX 75390-9039, USA

Corresponding author: Shay, Jerry W (Jerry.Shay@UTSouthwestern.edu)

Current Opinion in Cell Biology 2018, **52**:1–7

This review comes from a themed issue on **Cell nucleus**

Edited by **Mary C Dasso** and **Beatriz MA Fontoura**

https://doi.org/10.1016/j.ceb.2017.12.001

0955-0674/© 2017 Published by Elsevier Ltd.

Introduction

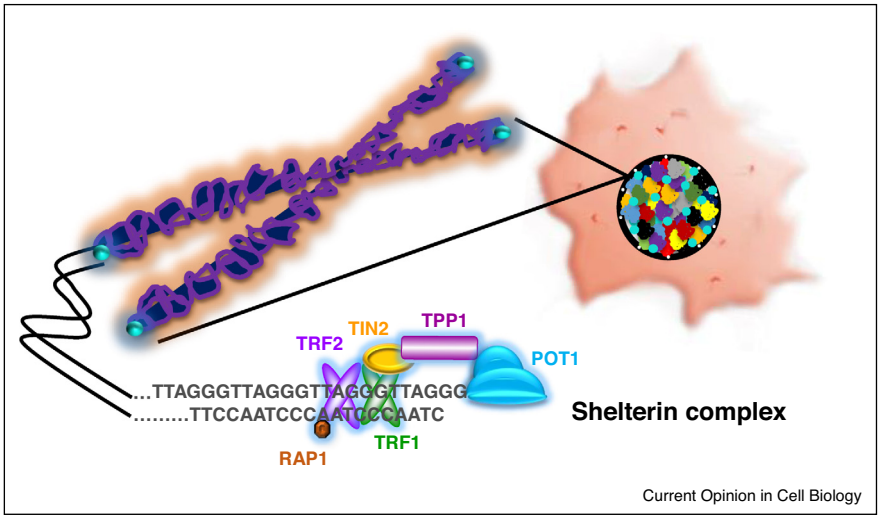
Three-dimensional (3D) chromatin looping mediate long-range interactions that bring distal genomic segments into close spatial proximity between enhancers and promoters that are linked to the regulation of gene expression [1•,2–4]. There are both cell type-specific and constitutive classes of genes that are regulated by 3D chromatin looping interactions. The ability to fold the entire human genome into the small volume of a typical nucleus involves occupancy of proteins such as CTCF (CCCTC-binding factor) and cohesion to form and maintain the chromatin loops resulting in insulated neighborhoods that are thought to protect genes from aberrant gene activation [1•,2–4]. However, the genome-wide landscape of long-range looping interactions and their dynamic properties across most lineages in humans remain largely uncharacterized. How progenitor cells differentiate through coordinated expression and repression of complex chromatin interactions remains an area of intense study. Even less well studied are the ends of linear chromosomes that are capped by thousands of hexameric telomeric $(TTAGGG)_n$ repeats [5] that also form 3D looping interactions. These types of interactions are particularly interesting since telomeres progressively shorten with increased aging and thus may provide a 'clocking' mechanism for gene regulation over long periods of time without initiating a DNA damage signaling response. A better understanding of the topological chromatin interactions at telomeres may provide new insights into how the genome and epigenome act in concert to regulate cell fate determinations during early human development and later in life as part of normal aging and disease progression.

Telomeres

Mammalian telomeres are repetitive DNA sequences, associated with the multimeric protein shelterin complex that facilitates formation of a lariat-like structure to shield the exposed ends of telomeric DNA from the DNA damage machinery [5,6]. The six member shelterin protein complex (Figure 1) includes TRF1 (telomeric repeat-binding factor 1) that binds to the canonical TTAGGG double-stranded telomeric repeats and interacts with TIN2 (TRF1-interaction nuclear factor 2). Another shelterin protein is TRF2 that also binds to double-stranded telomeric repeats and interacts with RAP1 (repressor/activator protein 1). POT1 (protection of telomeres 1) binds to single-stranded TTAGGG repeats and connects to TRF1 and TRF2 though a binding partner, TPP1, that also associates with TIN2. Information on how the complex of shelterin proteins maintain telomeres and preserve genomic integrity was recently reviewed [7•].

Due to incomplete lagging strand DNA synthesis, oxidative damage and other factors, progressive telomere shortening eventually results in cellular growth arrest and has been proposed as an initial proliferative barrier to tumor formation [8,9]. However, when the growth arrest initiated by a few shortened telomeres is bypassed due to loss of tumor suppressors and oncogenic changes, telomeres can continue to shorten eventually leading to chromosome bridge-fusion-breakage cycles that can result in genomic instability and an increased risk of cancer [9,10]. Eventually a mechanism is engaged to maintain telomeres as part of cancer development and this is almost universally accomplished by the reactivation of the telomerase reverse transcriptase (*TERT*) that produces

Figure 1



Human telomeres are repetitive DNA sequences at the ends of linear chromosomes. Telomeres function in nuclear architecture, chromosome positioning, and are often associated with the nuclear envelope. Telomeres also protect the linear ends of chromosomes from being recognized as DNA damage needing repair which occurs by interactions with the complex of six shelterin proteins. The very end of telomeric DNA consists of a single strand G-rich overhang that loops back and strand invades into the duplex TTAGGG repeats forming a T-loop. However, during replication the telomeres must unfold so the free end is accessible to the replication machinery. In total the telomere ends only account for ~1/6000th of the total genomic DNA in a cell. Telomeres progressively shorten with each cell division in the absence of a telomere maintenance mechanism eventually leading to a proliferation arrest that can be considered an initial potent tumor suppressor mechanism in large and long-lived mammals. However, loss of telomere protection can lead to telomere crisis and rare cells that escape crisis can promote cancer progression by almost universally activating telomerase.

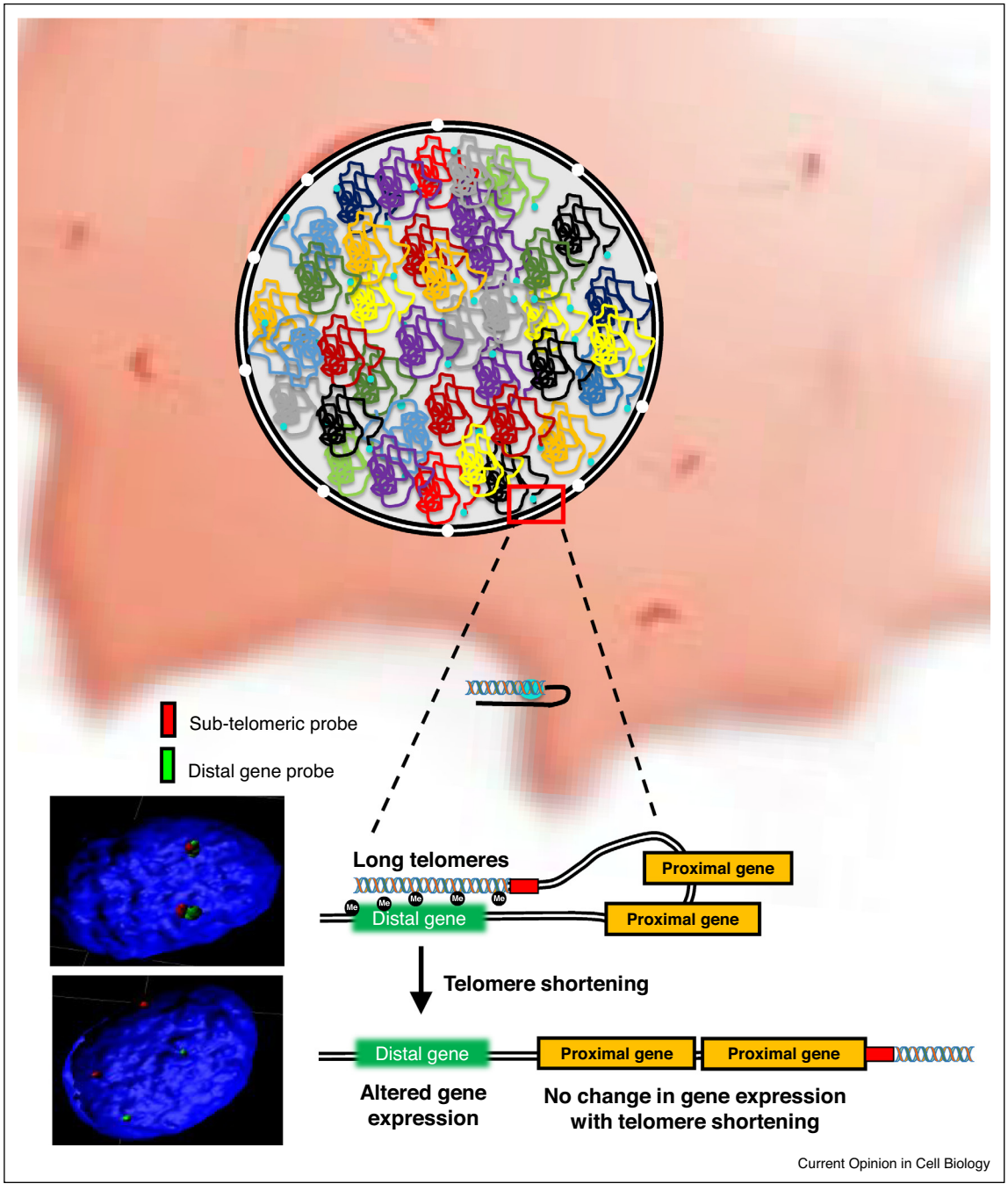
telomerase enzymatic activity when associated with other essential components to form a ribonucleoprotein complex [11,12]. Telomerase synthesizes telomeric DNA using the integral or template telomerase RNA component (*TERC*). During early human development telomerase is active but becomes silenced between 12 and 18 weeks of gestation [13,14]. There is evidence that *TERT* alternative splicing may be involved in silencing telomerase [13–15] thus limiting the maximal length of human telomeres. In addition to alternative splicing other mechanisms such epigenetic changes involving telomere 3D looping may also be involved.

Mechanisms regulating telomere length and telomerase

There are heterochromatin-associated histone marks (H3K9 and H4K20) associated with nucleosomes at the canonical TTAGGG telomeres and the non-canonical subtelomeric degenerative repeats that are similar but not identical to the histone marks present in constituitive heterochromatin [16]. Thus, telomeres have more compacted nucleosomes (altered spacing) potentially silencing genes in the vicinity of the telomeres. Similar to yeast and fly chromosome ends, mammalian telomeres can transcriptionally silence nearby genes through a process known as TPE (Telomere Position Effects) supporting the concept there are multiple epigenetic modifications at mammalian telomeres that organize them into a unique chromatin structure [16,17]. TPE in humans is influenced by telomere length that involves histone hypoacetylation that can be disrupted by treatment with deacetylase inhibitors [17]. Thus, the length of telomere repeats may influence heterochromatin assembly at telomeres and in part regulate gene expression changes. The first human disease in which the TPE model of continuous heterochromatin spreading that declines with distance from the telomere was FSHD (facioscapulohumeral muscular dystrophy) [18]. In another study the *ISG15* gene located 1 megabase from the end of chromosome 1p was reported to be regulated by telomere length [19]. Because genes between *ISG15* and the telomere were not regulated by telomere length a different mechanism of gene regulation by telomeres was clearly involved.

Recently, telomere 3D organization has been implicated in cellular processes in addition to the well-established telomere role in end protection (Figure 2). Telomeres are also involved in a process known as interstitial telomere loops (ITLs) or telomere position effects over long distances (TPE-OLD) [20,21,22••,23,24••,25]. Understanding the mechanisms that regulate the maximal length of human telomeres during human embryonic development and the mechanisms of how progressive telomere shortening leads to epigenetic alterations and changes in gene expression associated with human aging and pathologies is a current area of interest.

Figure 2

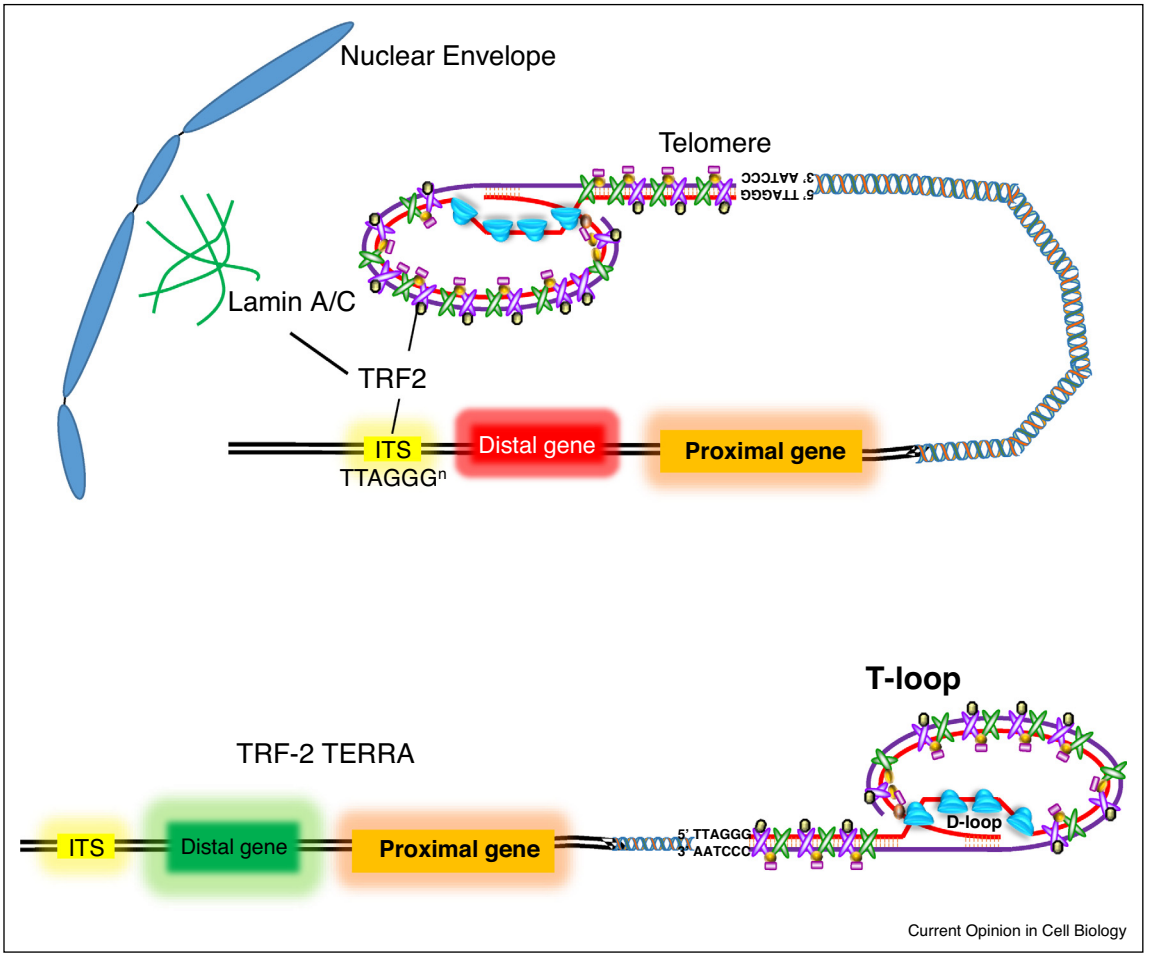


Telomere looping. In human development when telomeres get sufficient long, telomere looping can silence the *TERT* locus thus limiting the maximal length of human telomeres. Most adult somatic cells do not express telomerase activity and thus lose telomere length with each division. When telomeres get sufficiently short telomere looping at the *TERT* locus is altered. Using 3D-FISH to spatially localize a subtelomere probe (red) with a distal gene probe (green), cells with long telomeres often show co-localization but as telomeres progressively shorten the subtelomeric probe and the distal gene probe of interest separate. This separation from the telomere region correlates with altered gene expression at the distal gene regulated by telomere position effects over long distances (TPE-OLD).

Laminopathies and telomeropathies

The nuclear envelope consists of the outer membrane, the inner nuclear membrane and associated nuclear lamina, and the nuclear pore complex (Figure 3). There are several studies connecting telomeres and lamins (a scaffold of intermediate filament proteins associated with the inner membrane of the mammalian nucleus). Lamins directly interact with DNA and histones, and thus

Figure 3



Model of interactions between a telomere and a gene distal from the telomere. When cells have long telomeres 3D chromosomal looping positions a distal gene close to a telomere through interactions with interstitial telomere sequences (ITS) that is facilitated by shelterin proteins (such as TRF2) and the proteins that associate with the inner nuclear envelope (Lamin A/C). 3D telomere looping changes the chromatin structure near the distal gene, resulting in a change in gene expression. When cells have short telomeres these associations are lost and the expression of the distal gene changes. There is also support for telomere repeat-containing RNA (TERRA) that is a long noncoding RNA transcribed from telomeres that increases in cells with short telomeres and TERRA may sequester TRF2 from the distal telomere length regulated gene.

contribute to the higher-order chromatin organization in the nucleus [25,26•]. Alterations in lamin proteins affect nuclear architecture, genome topology as well as gene expression, and have been associated with human pathologies [26•]. The nuclear lamina consist of A-type, B-type and C-type lamins that also include alternative splice variants [27]. Mammalian chromatin interacts with the nuclear lamina through numerous lamina-associated domains (LAD). LADs help organize chromosomes and are generally associated with gene repression [25,27]. The anchoring of LADs to the nuclear lamina are believed to contribute to the overall folding of interphase chromosomes [26•]. Hutchison-Gilford Progeria Syndrome (HGPS), a premature aging syndrome, is caused by a point mutation in the *LMNA* gene, resulting in alternative splicing of lamin A to produce expression of a truncated protein termed progerin [28]. Progerin accumulation at the nuclear envelope disrupts nuclear architecture and chromatin organization leading to multiple severe pathologies. There is a correlation between HGPS and accelerated telomere shortening [29–31,32•] leading to DNA damage responses. Progressive telomere shortening in some humans without HGPS may also activate progerin [31], thus connecting lamins and telomeres with both normal and pathological aging.

Telomere maintenance spectrum disorders (telomeropathies) such as dyskeratosis congenita and familial idiopathic pulmonary fibrosis are genetically inherited pathologies and often show an early age of onset and genetic anticipation in future generations [33]. Mutations in a variety of genes associated with telomere and telomerase function, such as *TERT*, *TERC*, *PARN*, *RTEL1*, *TIN2*, *DKC1*, *NHP2*, *NOP10*, *Apollo*, *TCAB1* and *CTC1* are likely

to be only a subset of alterations since the network of genes associated with telomere maintenance is very large [33]. These human disorders are often associated with inherited telomere lengths being in the lower 1–10% range compared to age matched controls leading to premature telomere shortening and a spectrum of diseases. Several mechanisms of how shortened telomeres lead to differences in age of onset and symptoms of telomere disorders have been suggested [33].

Telomere position effect mechanisms

Age-dependent telomere erosion may be important in modulating the expression of genes adjacent (classic TPE) or at long distances from the telomeres (TPE-OLD or telomere looping, TL). In classic TPE, the degree of repression declines with distance from the telomeres [17–19]. One would expect in classic TPE that genes closest to the telomeres would be de-repressed first and then in a continuous fashion the next adjacent gene would be altered and these genes would be regulated by changes in chromatin. Using reporter assays, classic TPE was demonstrated in human cells by showing that a reporter gene located in non-telomeric regions was highly expressed but greatly downregulated when positioned at a telomere [17]. The first human gene to be regulated by TPE was reported in 2009 [19]. This study demonstrated that interferon stimulated gene 15 (ISG15) expressed at low levels in young cells with long telomeres, showed increased expression as telomeres progressively shortened. To document that expression of ISG15 was not due to DNA damage signaling from a short telomere, telomeres in old cells were experimentally elongated by ectopically introducing *TERT* [19]. Telomerase preferentially elongates the shortest telomeres and while no changes in ISG15 expression occurred during the first few cell divisions after *TERT* introduction (which would have eliminated any DNA damage signaling from a too short telomere), when telomeres elongated sufficiently (~50 cell divisions), ISG15 was again silenced. However, when genes between ISG15 and the telomere were examined, there was no changes in gene expression that correlated with telomere length. Thus, the phenomenon of TPE-OLD (Telomere Position Effect Over Long Distances (TPE-OLD) was demonstrated for the first time [20].

These studies showed that long telomeres could physically associate with genomic DNA many megabases from the telomere by 3D looping interactions. Independently, it was shown that telomeres could loop to interact at sites known as interstitial telomeric sequences (ITS) [34•] through lamin A/C and TRF2 [22••,23]. With whole genome sequencing it is now known that the human genome contains 1000s of $TTAGGG_n$ repeats (some of which can be quite long) not only at chromosome ends but throughout the genome [22••,33]. Thus, interstitial telomere loops (ITLs) via TRF2 and lamin A/C proteins could form stable associations far from the telomere ends (Figure 3). These studies offer a clue and molecular mechanism into how a laminopathy such as HGPS could be associated with telomere shortening and a premature aging syndrome. Since TRF2 interacts with lamin A/C but progerin does not, the mutation in HGPS could disrupt the normal homeostasis and stability of telomeres. While these studies connecting ITL formation with distal genes did not directly consider progressive telomeres shortening with gene regulation [22••], the studies on TPE-OLD did [20] and demonstrated progressive telomeres shortening was accompanied by changes in gene expression [20,24••].

While unprotected telomeres induce DNA damage signals leading to senescence, another equally plausible model is that progressive telomere shortening leads to changes in the expression of genes far from the telomeres and this may initiate age-related pathologies such as cancer without necessarily invoking a DNA damage signal. In humans and most large long-lived mammals, but not rodents, the *TERT* locus is very close to a telomere. Chromosome looping occurs between the human 5p subtelomere and ITS near the *CLPTM1L/TERT* locus resulting in a repressed epigenetic state when telomeres are long that is altered when telomeres shorten with increased age [24••]. While telomerase activity is detected during early human fetal development, it becomes silenced in most tissues, perhaps to prevent the early onset of cancer in large/long-lived mammals. Thus, the *hTERT* gene may autoregulate itself through telomere looping as an example of antagonist pleiotropy, where a gene that may be beneficial to the organism fitness early in life (by silencing telomerase and preventing the early onset of cancer), and the same gene is detrimental to the organism fitness post-reproduction late in life (telomerase reactivation in cancer when telomeres are short). Thus, the spatio-temporal expression of telomerase is tightly regulated in human cells.

Conclusions

Telomere looping is associated with interactions of the heterochromatic telomeric DNA with normal chromatin at distal sites potentially through interactions of the nuclear lamin A/C proteins, TRF2 and interstitial telomeric sequences (ITS) [35–38]. While there is substantial research into understanding how general chromosome 3D looping affects the regulation of gene expression, we have just begun to explore the phenomenon of long-range chromosome-end looping. Importantly, changes in telomere looping with increased age (and progressive telomere shortening) may be one mechanism of how cells time changes in physiology over decades without initiating DNA damage responses. A better understanding of the role of telomere dynamics in laminopathies and telomeropathies may lead to novel approaches to

intervene and perhaps prevent or reverse some of the consequences of these genetic diseases.

Acknowledgements

This work was supported by the National Institutes of Health [AG01228], the Harold Simmons National Cancer Institute Designated Comprehensive Cancer Center Support Grant [CA142543]; and the Southland Financial Corporation Distinguished Chair in Geriatric Research. This work was performed in laboratories constructed with support from National Institutes of Health [C06 RR30414].

References and recommended reading

Papers of particular interest, published within the period of review, have been highlighted as:

- • of special interest
- •• of outstanding interest

1. • Rowley MJ, Corces VG: **The three-dimensional genome: principles and roles of long-distance interactions**. *Curr Opin Cell Biol* 2016, **40**:8-14.

Review covers the mechanisms by which long distance chromatin interactions determine the 3D organization of the genome.

2. Dekker J, Mirny L: **The 3D genome as moderator of chromosomal communication**. *Cell* 2016, **164**:1110-1120.

3. Canela A, Maman Y, Jung S, Wong N, Callen E, Day A, Kieffer-Kwon K-R, Pekowska A, Zhang H, Rao SSP, Huan S-C *et al.*: **Genome organization drives chromosome fragility**. *Cell* 2017, **170**:507-521.

4. Chandra T, Kirschner K: **Chromosome organization during ageing and senescence**. *Curr Opin Cell Biol* 2016, **40**:161-167.

5. Greider CW: **Telomeres**. *Curr Opin Cell Biol* 1991, **3**:444-451.

6. de Lange T: **Shelterin: the protein complex that shapes and safeguards human telomeres**. *Genes Dev* 2005, **19**:2100-2110.

7. • Maciejowski J, de Lange T: **Telomeres in cancer: tumour suppression and genome instability**. *Nat Rev Mol Cell Biol* 2017, **18**:175-186.

Review on how short telomeres may have a tumor suppression effect but loss of telomere protection can lead to telomere crisis, genome instability and promotion of cancer.

8. Shay JW: **Role of telomeres and telomerase in aging and cancer**. *Cancer Discov* 2016, **6**:584-593.

9. Shay J, Wright WE, Werbin H: **Defining the molecular mechanisms of human cell immortalization**. *Biochim Biophys Acta* 1991, **1072**:1-7.

10. Wright WE, Pereira-Smith OM, Shay JW: **Reversible cellular senescence: a two-stage model for the immortalization of normal human diploid fibroblasts**. *Mol Cell Biol* 1989, **9**:3088-3092.

11. Shay JW, Bacchetti S: **A survey of telomerase activity in human cancer**. *Eur J Cancer* 1997, **33**:787-791.

12. Kim NW, Piatyszek MA, Prowse KR, Harley CB, West MD, Ho PLC, Coviello GM, Wright WE, Weinrich SL, Shay JW: **Specific association of human telomerase activity with immortal cells and cancer**. *Science* 1994, **266**:2011-2015.

13. Wright WE, Piatyszek MA, Rainey WE, Byrd W, Shay JW: **Telomerase activity in human germline and embryonic tissues and cells**. *Dev Genet* 1996, **18**:173-179.

14. Ulaner GA, Hu JF, Vu TH, Giudice LC, Hoffman AR: **Tissue-specific alternate splicing of human telomerase reverse transcriptase (hTERT) influences telomere lengths during human development**. *Int J Cancer* 2001, **91**:644-649.

15. Wong MS, Wright WE, Shay JW: **Alternative splicing regulation of telomerase: a new paradigm?** *Trends Genet* 2014, **30**:430-438.

16. Blasco MA: **The epigenetic regulation of mammalian telomeres**. *Nat Rev Genet* 2007, **8**:299-309.

17. Baur JA, Zou Y, Shay JW, Wright WE: **Telomere position effect in human cells**. *Science* 2001, **292**:2075-2077.

18. Stadler G, Rahimov F, King OD, Chen J, Robin JD, Wagner KR, Shay JW, Emerson CP, Wright WE: **Human disease and telomere position effect (TPE): the regulation of DUX4 expression in facioscapulohumeral muscular dystrophy (FSHD)**. *Nat Struct Mol Biol* 2013, **20**:671-678.

19. Lou Z, Jun W, Riethman H, Baur JA, Voglauer R, Shay JW, Wright WE: **Endogenous genes near telomeres regulated by telomere length in human cells**. *Aging* 2009, **1**:608-621.

20. Robin JD, Ludlow AT, Chen M, Magdinier F, Batten K, Holohan B, Stadler G, Wagner KR, Shay JW, Wright WE: **Telomere position effect: regulation of gene expression with progressive telomere shortening over long distances**. *Genes Dev* 2014, **28**:2464-2476.

21. Robin JD, Ludlow AT, Batten K, Gaillard M-C, Stadler G, Magdinier F, Wright WE, Shay JW: **SORBS2 transcription is activated by telomere position effect-over long distance upon telomere shortening in muscle cells from patients with facioscapulohumeral dystrophy**. *Genome Res* 2015, **25**:1781-1790.

22. •• Wood AM, Laster K, Rice EI, Kosak ST: **A beginning of the end: new insights into the functional organization of telomeres**. *Nucleus* 2015, **6**:172-178.

A review into the relationship between Lamin A/C and telomere protection with a discussion of interstitial telomere loops (ITL) and the regulation of gene expression at sites distal from the telomere.

23. Wood AM, Rendtlew Danielsen JM, Lucas CA, Rice EL, Scalzo D, Shimi T, Goldman RD, Smith ED, LeBrau MM, Kosak ST: **TRF2 and lamin A/C interact to facilitate the functional organization of chromosome ends**. *Nat Commun* 2014, **5**:5467 http://dx.doi.org/10.1038/ncomms6467.

24. •• Kim W, Ludlow A, Min J, Stadler G, Robin JD, Zhang N, Mender I, Wright WE, Shay JW: **Regulation of the human telomerase gene *TERT* by telomere position effect over long distance (TPE-OLD): implications for aging and cancer**. *PLoS Biol* 2016, **14** http://dx.doi.org/10.1371/journal.pbio.2000016.

The human *TERT* gene is regulated by telomere position effects over long distances (TPE-OLD) or interstitial telomere looping (ITL). Provides an explanation on how maximal telomere length is determined during human development with the silencing of *TERT*. With increased age shortened telomeres lead to the *TERT* locus becoming permissive for transcription and this often correlates with *TERT* promoter mutations and cancer progression.

25. Robin JD, Magdinier F: **Physiological and pathological aging affects chromatin dynamics, structure and function at the nuclear edge**. *Front Genet* 2016, **7** http://dx.doi.org/10.3389/fgene.2016.00153.

26. • Dorado B, Andres V: **A-type lamins and cardiovascular disease in premature aging syndromes**. *Curr Opin Cell Biol* 2017, **46**:17-25.

Review on the role of Lamin A and its mutant form (progerin) and the onset of laminopathies including cardiovascular disease.

27. van Steensel B, Belmont AS: **Lamina-associated domains: links with chromosome architecture, heterochromatin, and gene repression**. *Cell* 2017, **169**:780-791.

28. Gordon LB, Rothman FG, López-Otín C, Misteli T: **Progeria: a paradigm for translational medicine**. *Cell* 2014, **156**:400-407.

29. McCord RP, Nazario-Toole A, Zhang H, Chines PS, Zhan Y, Erdos MR, Collins FS, Dekker J, Cao K: **Correlated alterations in genome organization, histone methylation, and DNA-lamin A/C interactions in Hutchinson–Gilford progeria syndrome**. *Genome Res* 2013, **23**:260-269.

30. Chojnowski A, Ong PF, Wong ESM, Lim JSY, Mutalif RA, Navasankari R, Dutta B, Yang H, Liow YY, Sze SK *et al.*: **Progerin reduces LAP2α-telomere association in Hutchinson–Gilford progeria**. *Elife* 2015:4.

31. Cao K, Blair CD, Faddah DA, Kieckhaefer JE, Olive M, Erdos MR, Nabel EG, Collins FS: **Progerin and telomere dysfunction collaborate to trigger cellular senescence in normal human fibroblasts**. *J Clin Invest* 2011, **121**:2833-2844.

32. • Burla R, La Torre M, Saggio I: **Mammalian telomeres and their partnership with lamins**. *Nucleus* 2016, **7**:187-202.
Review on the physical connections of telomeres and lamins (a nuclear envelope and matrix component) and implications in cell senescence.

33. Holohan B, Wright WE, Shay JW: **Impaired telomere maintenance spectrum disorders**. *J Cell Biol* 2014, **205**:289-299.

34. • Bolzan AD: **Interstitial telomeric sequences in vertebrate chromosomes: origin, function, instability, and evolution**. *Mutat Res* 2017, **773**:51-65.
Review on the different types of interstitial telomeric sequences (ITS) including relationships between telomeres at the ends of chromosomes and ITS.

35. Hussain T, Saha D, Purohi G, Kra A, Muhkerjee AK, Sharma S, Sengupta S, Dhapola P, Maji B, Vedagrpuram S *et al.*: **Transcription regulation of CDKN1A (p21/CIP1/WAF1) by TRF2 is epigenetically controlled through the REST repressor complex**. *Sci Rep* 2017, **7**:11541 http://dx.doi.org/10.1038/s41598-017-11177-1.

36. Saha D, Singh A, Hussain T, Srivastava V, Sengupta S, Kar A, Dhapola P, Dhople V, Ummani R, Chowdhury S: **Epigenetic suppression of human telomerase (hTERT) is mediated by the metastasis suppressor NME2 in a G-quadruplex-dependent fashion**. *J Biol Chem* 2017. manuscript M117.792077, in press.

37. Simonet T, Laurew-Emanuelle Z, Philippe C, Lebrigand K, Schouteden C, Augereau A, Bauwens S, Ye J, Santagostino M, Giulotto E *et al.*: **The human TTAGGG repeat factors 1 and 2 bind to a subset of interstitial telomeric sequences and satellite repeats**. *Cell Res* 2011, **21**:1028-1038 http://dx.doi.org/10.1038/cr.2011.40.

38. Min J, Shay JW: ***TERT* promoter mutations enhance telomerase activation by long-range chromatin interactions**. *Cancer Discov* 2016, **6**:1212-1214.